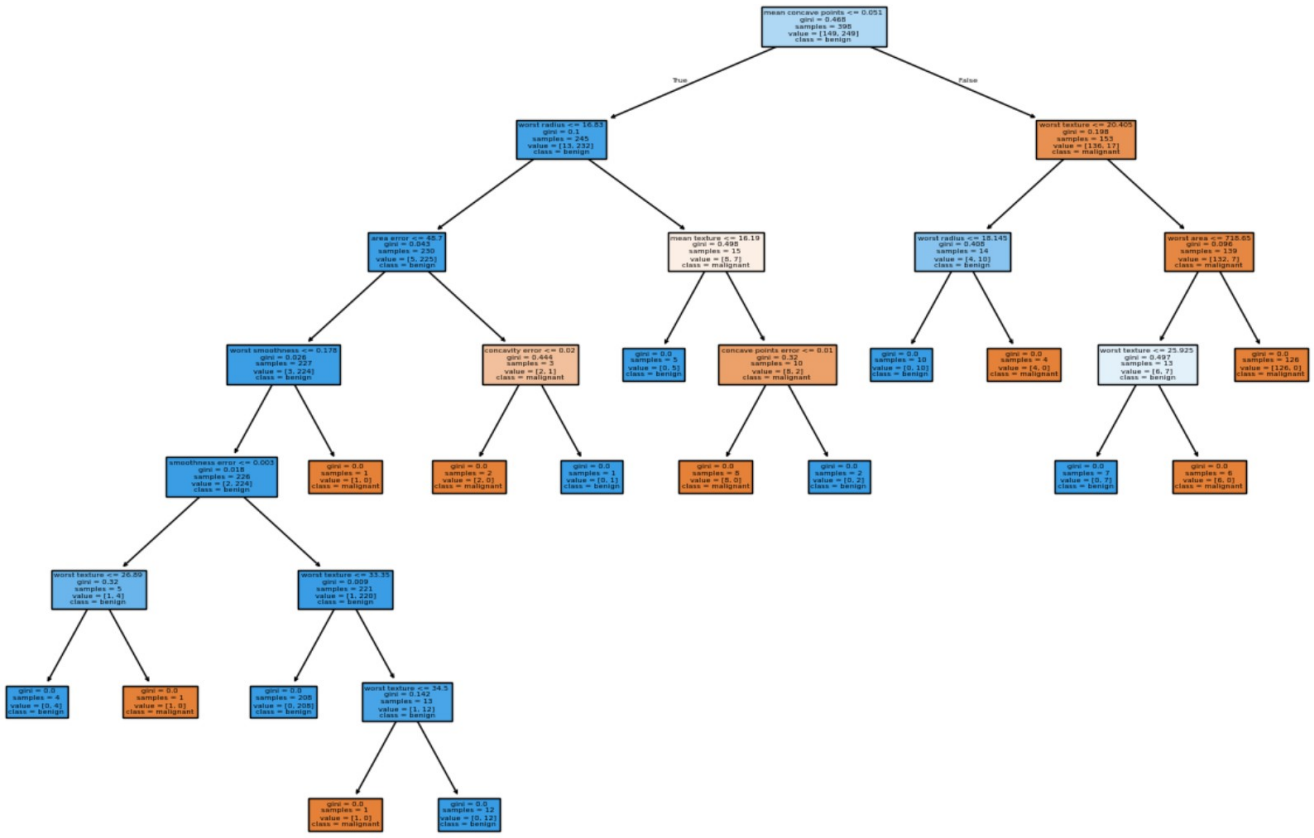
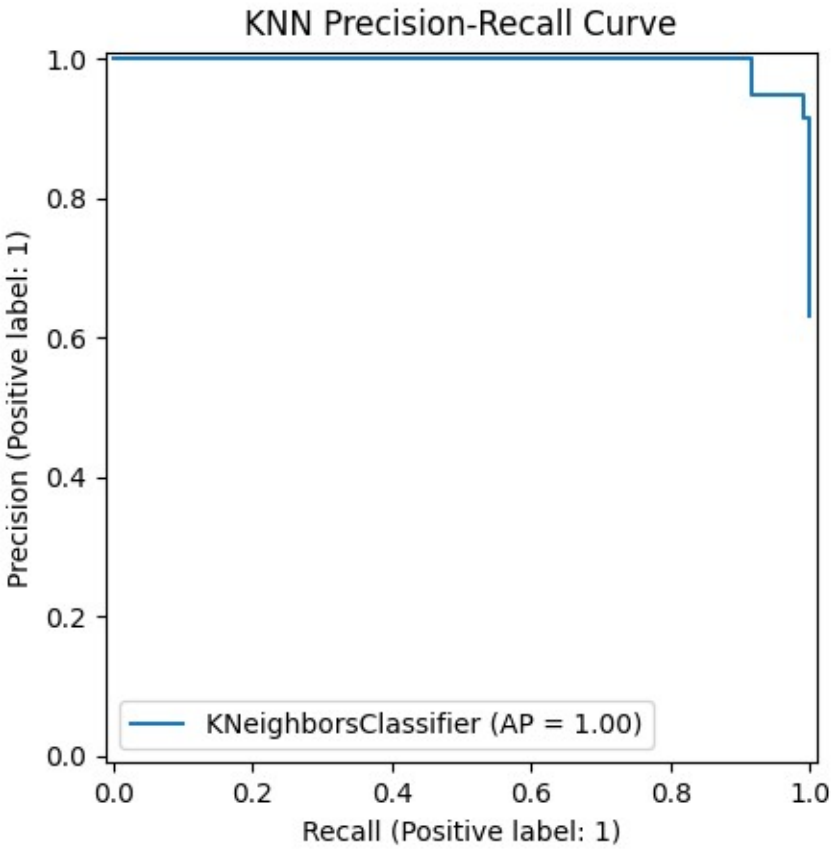
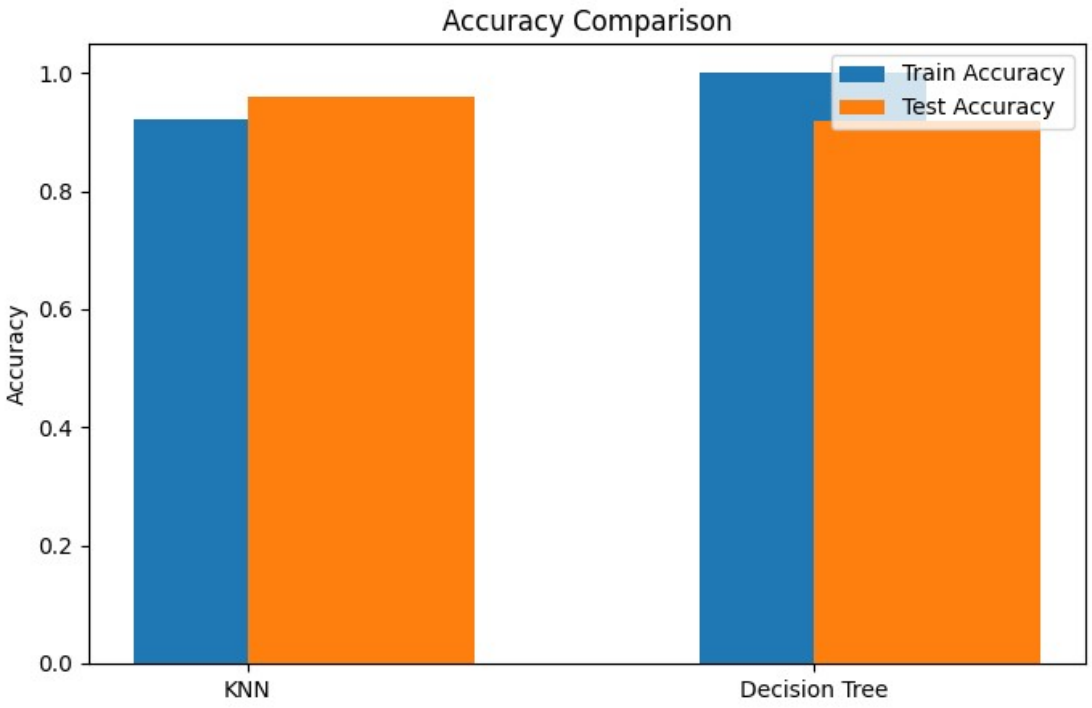
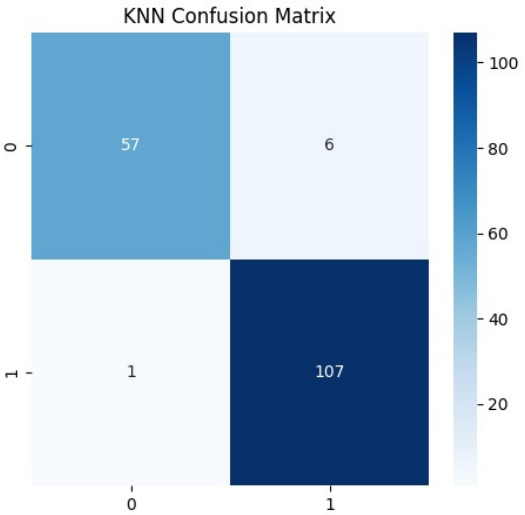
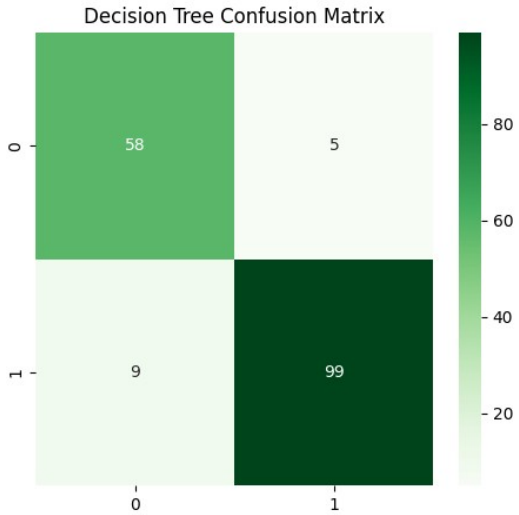
```
❯ python3 l6_e1.py && python3 l6_e2.py && python3 l6_e3.py
--- Experiment 1: KNN Results ---
Accuracy: 0.9591

Confusion Matrix:
 [[ 57   6]
 [  1 107]]

Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.90      0.94        63
           1       0.95      0.99      0.97       108

    accuracy                           0.96       171
   macro avg       0.96      0.95      0.96       171
weighted avg       0.96      0.96      0.96       171

Plot saved as knn_precision_recall.png
--- Experiment 2: Decision Tree Results ---
Accuracy: 0.9415

Confusion Matrix:
 [[ 60   3]
 [  7 101]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.95      0.92        63
           1       0.97      0.94      0.95       108

    accuracy                           0.94       171
   macro avg       0.93      0.94      0.94       171
weighted avg       0.94      0.94      0.94       171

Plots saved as dtree_precision_recall.png and decision_tree_visu
alization.png
Comparison plots saved.
```

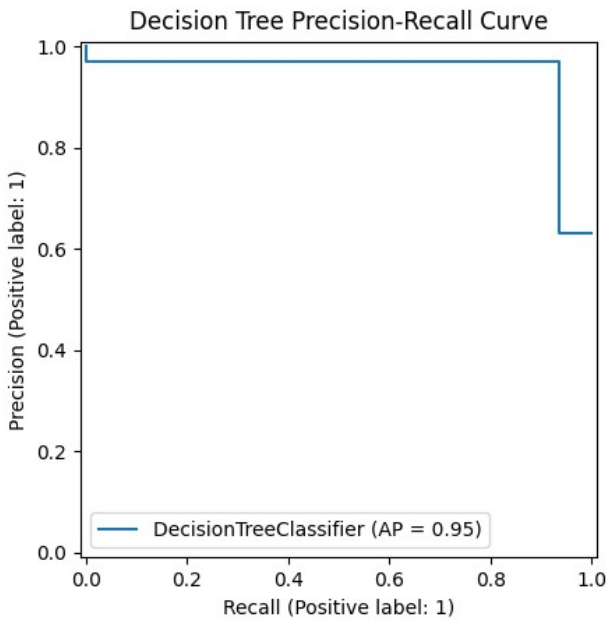

mean concave points <= 0.051
gini = 0.468
samples = 398
value = [149, 249]
class = benign
True
False
worst radius <= 16.83
gini = 0.1
samples = 245
value = [13, 232]
class = benign
worst texture <= 20.405
gini = 0.198
samples = 153
value = [136, 17]
class = malignant
area error <= 48.7
gini = 0.043
samples = 230
value = [5, 225]
class = benign
mean texture <= 16.19
gini = 0.498
samples = 15
value = [8, 7]
class = malignant
worst radius <= 18.145
gini = 0.408
samples = 14
value = [4, 10]
class = benign
worst area <= 718.65
gini = 0.096
samples = 139
value = [132, 7]
class = malignant
worst smoothness <= 0.178
gini = 0.026
samples = 227
value = [3, 224]
class = benign
concavity error <= 0.02
gini = 0.444
samples = 3
value = [2, 1]
class = malignant
gini = 0.0
samples = 5
value = [0, 5]
class = benign
concave points error <= 0.01
gini = 0.32
samples = 10
value = [8, 2]
class = malignant
gini = 0.0
samples = 10
value = [0, 10]
class = benign
gini = 0.0
samples = 4
value = [4, 0]
class = malignant
worst texture <= 25.925
gini = 0.497
samples = 13
value = [6, 7]
class = benign
gini = 0.0
samples = 126
value = [126, 0]
class = malignant
smoothness error <= 0.003
gini = 0.018
samples = 226
value = [2, 224]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 2
value = [2, 0]
class = malignant
gini = 0.0
samples = 1
value = [0, 1]
class = benign
gini = 0.0
samples = 8
value = [8, 0]
class = malignant
gini = 0.0
samples = 2
value = [0, 2]
class = benign
gini = 0.0
samples = 7
value = [0, 7]
class = benign
gini = 0.0
samples = 6
value = [6, 0]
class = malignant
worst texture <= 26.89
gini = 0.32
samples = 5
value = [1, 4]
class = benign
worst texture <= 33.35
gini = 0.009
samples = 221
value = [1, 220]
class = benign
gini = 0.0
samples = 4
value = [0, 4]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 208
value = [0, 208]
class = benign
worst texture <= 34.5
gini = 0.142
samples = 13
value = [1, 12]
class = benign
gini = 0.0
samples = 1
value = [1, 0]
class = malignant
gini = 0.0
samples = 12
value = [0, 12]
class = benign


KNN Precision-Recall Curve
Precision (Positive label: 1)
Recall (Positive label: 1)
KNeighborsClassifier (AP = 1.00)
0.0
0.2
0.4
0.6
0.8
1.0


Accuracy Comparison
Train Accuracy
Test Accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Decision Tree


KNN Confusion Matrix
57
6
1
107
0
1
100
80
60
40
20


Decision Tree Confusion Matrix
58
5
9
99
0
1
80
60
40
20


Decision Tree Precision-Recall Curve
Precision (Positive label: 1)
Recall (Positive label: 1)
DecisionTreeClassifier (AP = 0.95)
1.0
0.8
0.6
0.4
0.2
0.0